AWS Machine Learning Specialty (MLS-C01)

4-Hour Instructor-Led Exam Preparation Instructor Guide

CONFIDENTIAL — Instructor Copy — Not for Distribution

Table of Contents

Session Overview

4-Hour Agenda

Time	Duration	Section	Exam Weight
0:00 – 0:55	55 min	Domain 1: Data Engineering	20%
0:55 – 1:00	5 min	Break	—
1:00 – 2:00	60 min	Domain 2: Exploratory Data Analysis	24%
2:00 – 3:30	90 min	Domain 3: Modeling	36%
3:20 – 3:30	10 min	Break	—
3:30 – 4:00	55 min*	Domain 4: ML Implementation & Operations	20%
4:00 – 4:55	55 min	Final Review & Practice Exam	—

* Timing is approximate; adjust based on class pace. The 4-hour runtime excludes breaks.

Materials Needed

- This printed instructor guide
- Projector / screen for any live demos
- Whiteboard or flip chart for diagrams
- Timer (phone or watch) for section pacing
- Student handouts (optional — condensed cheat sheet version)
- Pens / highlighters for students

How to Use This Guide

Prefix / Format	Meaning
[EXAM TIP] (red)	Key exam knowledge — emphasize verbally and ensure students note it
[INSTRUCTOR NOTE] (blue)	Delivery guidance for you — do NOT read aloud
[LIFE SCIENCES EXAMPLE] (green)	Domain-specific scenario to walk through with the class
Bold text	Key terms, service names, concepts students must know
Italic text	Supplementary context or exam strategy advice

Prefix / Format	Meaning
Tables	Comparison content — walk through row by row

Domain 1: Data Engineering

Time Allocation: 55 minutes | Exam Weight: 20% | Questions: ~13 of 65

This domain covers how to create data repositories for ML and how to implement data ingestion and transformation solutions.

1.1 Data Ingestion & Collection (15 min)

[INSTRUCTOR NOTE] Start by asking: 'What are the two fundamental paradigms for getting data into AWS?' Guide toward batch vs streaming.

Batch vs. Streaming Ingestion

Characteristic	Batch Ingestion	Streaming Ingestion
Latency	Minutes to hours	Milliseconds to seconds
Data Volume per Job	Large (GB–TB)	Small records continuously
AWS Services	Glue, EMR, Data Pipeline, Batch	Kinesis, MSK (Kafka)
Use Cases	ETL, nightly reports, bulk loads	Real-time dashboards, alerts, live ML inference
Cost Model	Pay per job/cluster runtime	Pay per shard-hour or throughput
Exam Keyword	'periodically', 'nightly', 'bulk'	'real-time', 'streaming', 'continuous'

Amazon Kinesis Family

Feature	Kinesis Data Streams (KDS)	Kinesis Data Firehose	Kinesis Data Analytics
Purpose	Real-time data streaming	Delivery to destinations	Real-time SQL/Flink analytics
Provisioning	Manual (shard count)	Fully managed / auto-scaling	Fully managed
Latency	~200ms	60s buffer (min)	Sub-second

Feature	Kinesis Data Streams (KDS)	Kinesis Data Firehose	Kinesis Data Analytics
Retention	24h default, up to 365 days	No retention (pass-through)	No retention
Consumers	Custom (KCL, Lambda, etc.)	S3, Redshift, OpenSearch, Splunk, HTTP	KDS or Firehose output
Transforms	None built-in	Lambda transform (optional)	SQL or Apache Flink
Ordering	Per-shard ordering	No ordering guarantee	Per-key (Flink)
Exam Keyword	'custom processing', 'multiple consumers', 'replay'	'delivery', 'near-real-time', 'load to S3/Redshift'	'SQL on streams', 'anomaly detection', 'real-time aggregation'

[EXAM TIP] *If the question says 'real-time' AND 'multiple consumers' or 'replay' → KDS. If it says 'deliver to S3/Redshift' with minimal code → Firehose. If it says 'run SQL on streaming data' → Kinesis Data Analytics.*

AWS Glue

Glue Component	Purpose	Key Detail
Glue Crawlers	Auto-discover schema from data in S3, RDS, etc.	Populates the Glue Data Catalog
Glue Data Catalog	Central metadata repository (Hive-compatible)	Used by Athena, EMR, Redshift Spectrum
Glue ETL Jobs	Serverless Spark-based ETL	Supports PySpark and Scala; auto-generates code
Glue DataBrew	Visual no-code data preparation	250+ built-in transforms; great for non-engineers
FindMatches ML	ML-based deduplication / record matching	No ML expertise needed; uses labeling
Glue Job Bookmarks	Track previously processed data	Prevents reprocessing; supports incremental loads

[EXAM TIP] *Glue is serverless ETL. If the question mentions 'serverless' + 'ETL' or 'transform' → think Glue first. If it mentions 'deduplicate' or 'match records' → Glue FindMatches ML Transform.*

Other Ingestion Services

Service	Use Case	Key Detail
AWS DMS	Database migration (ongoing replication)	Supports homogeneous and heterogeneous migrations; CDC
AWS Transfer Family	SFTP/FTPS/FTP to S3	Drop-in replacement for legacy file transfer servers
AWS DataSync	On-prem → AWS data transfer (agent-based)	Up to 10x faster than open-source tools; scheduled
AWS Snow Family	Offline bulk data transfer (physical devices)	Snowcone (8TB), Snowball Edge (80TB), Snowmobile (100PB)
AWS IoT Core	IoT device ingestion (MQTT)	Device gateway → Rules Engine → Kinesis/S3/Lambda

[LIFE SCIENCES EXAMPLE] A pharmaceutical company needs to ingest real-time clinical trial sensor data from 5,000 wearable devices monitoring patient vital signs. Each device sends heart rate, blood oxygen, and activity data every 5 seconds. Walk through: IoT Core → Kinesis Data Streams (real-time alerting on anomalous vitals) + Kinesis Firehose (batch delivery to S3 for later ML training). Discuss why Firehose alone wouldn't work here — they need the real-time consumer for patient safety alerts.

[LIFE SCIENCES EXAMPLE] A genomics lab generates 2 TB of sequencing data per run, with 3 runs per week. The data lives on an on-premises HPC cluster. Design the ingestion: DataSync for ongoing weekly transfers (2TB is feasible over a fast connection), or Snowball if they also need to migrate 200TB of historical sequence data.

[INSTRUCTOR NOTE] Discussion prompt: 'If you have both historical bulk data and ongoing new data, how do you handle both?' Answer: Snow Family for the bulk historical migration, DataSync or Kinesis for ongoing.

1.2 Data Storage & Transformation (15 min)

Amazon S3 as the ML Data Lake

S3 is the foundation of nearly every ML architecture on AWS. Key concepts:

Storage Class	Use Case	Retrieval	Cost (relative)
S3 Standard	Active training data, frequently accessed	Immediate	$$$$
S3 Intelligent-Tiering	Unknown access patterns	Immediate	$$$
S3 Standard-IA	Infrequent access (model artifacts archive)	Immediate	$$
S3 Glacier Instant	Rare access, instant retrieval	Immediate	$$

Storage Class	Use Case	Retrieval	Cost (relative)
S3 Glacier Flexible	Archival (old datasets)	Minutes to hours	$
S3 Glacier Deep Archive	Compliance / long-term archival	12–48 hours	$

[EXAM TIP] *Training data should be in S3 Standard. Old model artifacts or datasets you rarely retrain on can go to IA or Glacier. The exam tests whether you know retrieval times matter for ML pipelines.*

File Formats for ML

Format	Type	Columnar?	Best For	SageMaker Support
CSV	Text	No (row)	Small datasets, human-readable	Most algorithms
JSON / JSONL	Text	No	Semi-structured data, NLP	Some algorithms
Parquet	Binary	Yes	Large analytics datasets, Athena/Glue	Data Wrangler, Processing
ORC	Binary	Yes	Hive/EMR workloads	EMR-based training
RecordIO-Protobuf	Binary	No	SageMaker built-in algorithms (optimized)	Native; fastest for built-ins
TFRecord	Binary	No	TensorFlow models	TF containers on SageMaker
Image files (JPEG/PNG)	Binary	No	Computer vision	Image Classification, Object Detection

[EXAM TIP] *RecordIO-Protobuf is Amazon SageMaker's preferred format for built-in algorithms. It supports Pipe Mode (streaming from S3) for fastest training. If the exam asks about optimal format for SageMaker built-in algorithms → RecordIO-Protobuf.*

Data Lake Architecture (Zones)

Zone	Purpose	Format	Example
Raw / Landing	Original data as ingested	CSV, JSON, images as-is	Original clinical trial exports

Zone	Purpose	Format	Example
Curated / Clean	Cleaned, validated, standardized	Parquet (columnar, compressed)	De-identified, normalized records
ML-Ready / Feature	Feature-engineered, ML-formatted	RecordIO-Protobuf, CSV	Encoded features, train/val/test splits

Transformation Services Comparison

Service	Best For	Serverless?	Scale
AWS Glue ETL	General ETL, catalog integration	Yes	Auto-scaling Spark
Amazon EMR	Heavy Spark/Hadoop workloads, custom code	No (managed cluster)	Petabyte-scale
Amazon Athena	Ad-hoc SQL queries on S3 data	Yes	Per-query
Amazon Redshift	Data warehouse analytics, ML via Redshift ML	No (cluster)	Petabyte-scale
SageMaker Processing	ML-specific preprocessing	Yes (managed)	Configurable instances
SageMaker Data Wrangler	Visual data prep for ML	Yes	Built into Studio

[EXAM TIP] *Glue = serverless ETL. EMR = heavy custom Spark. Athena = ad-hoc SQL on S3. SageMaker Processing = ML-specific preprocessing (e.g., scikit-learn transforms). Know when to use each.*

[LIFE SCIENCES EXAMPLE] A biotech company stores protein structure data, gene expression matrices (CSV), and microscopy images (TIFF). Walk through storage decisions: gene expression data → convert to Parquet for Athena analytics, then RecordIO-Protobuf for SageMaker training. Microscopy images stay as-is in S3 for Image Classification. Protein data (structured) → Parquet in the curated zone. Emphasize the data lake zones: raw (original files), curated (cleaned, standardized), ML-ready (feature-engineered, formatted for algorithms).

[LIFE SCIENCES EXAMPLE] An EHR (Electronic Health Records) vendor needs to transform 50 million patient records for a readmission prediction model. They need to de-identify PHI, normalize lab values, and encode diagnosis codes. Walk through: Glue ETL for the bulk transform (serverless, HIPAA eligible), with a custom Lambda for PHI de-identification using Amazon Comprehend Medical to detect and redact medical entities.

1.3 Domain 1 Quiz (15 min)

[INSTRUCTOR NOTE] Read each question aloud. Give students 30 seconds to think, then reveal the answer. Explain WHY wrong answers are wrong.

Question 1:

A company needs to collect real-time clickstream data from its website and deliver it to S3 for later ML analysis. Which service is the MOST operationally efficient?

(A) Kinesis Data Streams with a custom S3 writer

(B) Kinesis Data Firehose

(C) Amazon SQS with a Lambda consumer writing to S3

(D) AWS Glue Streaming ETL

Correct Answer: (B)

Explanation: Firehose is purpose-built for delivery to S3/Redshift with zero custom code. KDS requires a custom consumer. SQS+Lambda works but is more complex. Glue Streaming is for ETL, not simple delivery.

Question 2:

A data engineer needs to migrate 150 TB of historical genomic data from on-premises to S3. The office has a 1 Gbps internet connection. What is the FASTEST approach?

(A) AWS DataSync over the internet

(B) S3 Transfer Acceleration

(C) AWS Snowball Edge devices

(D) AWS Direct Connect

Correct Answer: (C)

Explanation: At 1 Gbps, transferring 150TB would take ~14 days over the wire. Snowball Edge (80TB each, 2 devices) can be shipped in days. DataSync is for smaller ongoing transfers. Direct Connect takes weeks to provision.

Question 3:

Which AWS Glue feature should be used to prevent reprocessing of already-processed data in an incremental ETL pipeline?

(A) Glue Crawlers

(B) Glue FindMatches

(C) Glue Job Bookmarks

(D) Glue DataBrew

Correct Answer: (C)

Explanation: Job Bookmarks track which data has been processed so incremental jobs only process new data. Crawlers discover schema. FindMatches deduplicates. DataBrew is visual data prep.

Question 4:

A machine learning team wants to run SQL queries directly against data stored in S3 without loading it into a database. Which service should they use?

(A) Amazon Redshift

(B) Amazon RDS

(C) Amazon Athena

(D) Amazon DynamoDB

Correct Answer: (C)

Explanation: Athena is serverless interactive SQL that queries S3 directly. Redshift requires loading data into a cluster. RDS and DynamoDB are databases, not S3 query engines.

Question 5:

What is the recommended file format for FASTEST training performance with SageMaker built-in algorithms?

(A) CSV

(B) Parquet

(C) JSON

(D) RecordIO-Protobuf

Correct Answer: (D)

Explanation: RecordIO-Protobuf is optimized for SageMaker built-in algorithms and supports Pipe Mode for streaming directly from S3, avoiding disk I/O bottlenecks.

Question 6:

A company has 5,000 IoT sensors sending telemetry data via MQTT. They need real-time anomaly detection AND batch storage for historical analysis. Which architecture is correct?

(A) IoT Core → Kinesis Data Streams → Lambda (anomaly detection) + Kinesis Firehose → S3

(B) IoT Core → SQS → EC2 (anomaly detection) + S3

(C) IoT Core → Kinesis Firehose → S3 → Lambda (anomaly detection)

(D) IoT Core → DynamoDB → DynamoDB Streams → Lambda (anomaly detection)

Correct Answer: (A)

Explanation: IoT Core handles MQTT ingestion. KDS enables real-time processing with Lambda. Firehose delivers to S3 for batch. Option C puts anomaly detection after S3, adding latency. Option D is over-engineered.

Question 7:

Which Kinesis service would you use to run SQL queries on streaming data to detect anomalies in real-time?

(A) Kinesis Data Streams

(B) Kinesis Data Firehose

(C) Kinesis Data Analytics

(D) Kinesis Video Streams

Correct Answer: (C)

Explanation: Kinesis Data Analytics supports SQL and Apache Flink for real-time stream processing, including built-in anomaly detection functions. KDS is for ingestion, Firehose for delivery.

Question 8:

A data lake has three zones: raw, curated, and ML-ready. In which zone would you store Parquet files that have been cleaned and standardized but not yet feature-engineered?

(A) Raw zone

(B) Curated zone

(C) ML-ready zone

(D) Archive zone

Correct Answer: (B)

Explanation: The curated zone holds cleaned, validated, standardized data (typically Parquet). Raw holds original files as-is. ML-ready holds feature-engineered data formatted for algorithms.

BREAK — 5 minutes

> **[INSTRUCTOR NOTE]** Use this time to set up any materials for Domain 2. Check pacing — you should be at ~1:00 into the session.

Domain 2: Exploratory Data Analysis

Time Allocation: 60 minutes | Exam Weight: 24% | Questions: ~16 of 65

2.1 Data Visualization & Statistics (15 min)

Descriptive Statistics

Statistic	What It Measures	When It Matters for ML	Formula Hint
Mean	Central tendency (average)	Imputation, normalization baseline	Σx / n
Median	Central tendency (robust to outliers)	Better than mean for skewed data	Middle value

Statistic	What It Measures	When It Matters for ML	Formula Hint
Variance	Spread of data	Feature scaling decisions	$\Sigma(x-\mu)^2/n$
Standard Deviation	Spread (same units as data)	Standardization (z-score)	$\sqrt{\text{Variance}}$
Skewness	Asymmetry of distribution	Need for log/Box-Cox transform	>0 right-skewed, <0 left-skewed
Kurtosis	Tail heaviness	Outlier prevalence indicator	>3 heavy-tailed (leptokurtic)
Correlation	Linear relationship between variables	Feature selection, multicollinearity	Pearson r: -1 to +1

Visualization Types

Chart Type	Best For	ML Use Case
Histogram	Distribution shape of single variable	Detect skewness, choose transforms
Box Plot	Distribution + outliers + quartiles	Compare feature distributions across classes
Scatter Plot	Relationship between 2 variables	Spot correlations, clusters, outliers
Heatmap	Correlation matrix or confusion matrix	Feature selection, model evaluation
Line Chart	Time series trends	Seasonality detection, trend analysis
Bar Chart	Categorical comparisons	Class distribution, feature importance
Pair Plot	All pairwise relationships	Multi-feature exploration

Types of Data

Data Type	Examples	Encoding for ML
Numerical (Continuous)	Age, temperature, salary	Scaling (standardize/normalize)
Numerical (Discrete)	Count of events, number of children	Often used as-is or binned
Categorical (Nominal)	Color, country, diagnosis code	One-hot encoding

Data Type	Examples	Encoding for ML
Categorical (Ordinal)	Severity (low/med/high), education level	Label/ordinal encoding (preserves order)
Text	Clinical notes, reviews, descriptions	Tokenization, TF-IDF, word embeddings
Time Series	Stock prices, sensor readings, vital signs	Lag features, rolling statistics, decomposition

Statistical Distributions

Distribution	Shape	Example Use Case	Key Property
Normal (Gaussian)	Bell curve, symmetric	Heights, measurement errors	Mean = Median = Mode; 68-95-99.7 rule
Binomial	Discrete, symmetric around np	Pass/fail outcomes over n trials	n trials, probability p
Poisson	Discrete, right-skewed	Count of events in time window	Mean = Variance = λ
Uniform	Flat / rectangular	Random number generation	All values equally likely
Log-Normal	Right-skewed	Gene expression, income	Log of variable is normal

[LIFE SCIENCES EXAMPLE] You're analyzing a clinical trial dataset for a new cancer drug. Show the class how you'd explore it: histogram of patient ages (check for uniform enrollment), box plot of tumor sizes by treatment group (spot outliers — one patient's tumor measurement seems 10x larger, likely a data entry error in mm vs cm), scatter plot of dosage vs response rate, heatmap of correlation between biomarkers. Highlight that the tumor size outlier is exactly the kind of thing EDA catches before it ruins your model.

[LIFE SCIENCES EXAMPLE] Gene expression data typically follows a log-normal distribution (highly right-skewed). Ask students: what transform should we apply? (Log transform.) What type of visualization reveals this? (Histogram showing the skew, then histogram after log transform showing normality.) This connects skewness concepts to a real domain.

[EXAM TIP] *The exam loves asking about identifying outliers (box plots), detecting skewness (histograms), and understanding correlation vs causation. Know your visualization types cold.*

2.2 Feature Engineering (20 min)

[INSTRUCTOR NOTE] This is one of the highest-yield sections for the exam. Feature engineering questions appear across multiple domains.

Missing Data Strategies

Strategy	When to Use	Pros	Cons
Drop rows	< 5% missing, MCAR	Simple, no bias introduced	Lose data
Drop columns	Column is mostly missing (>50%)	Removes noise	Lose potentially useful feature
Mean/Median imputation	Numerical, MCAR/MAR	Simple, preserves sample size	Reduces variance, ignores relationships
Mode imputation	Categorical variables	Simple	Can introduce bias
KNN imputation	Complex patterns in data	Captures relationships between features	Computationally expensive
Indicator variable	Missingness is informative (MNAR)	Preserves signal in the missingness itself	Adds a feature column
Multiple imputation	Principled statistical analysis needed	Accounts for imputation uncertainty	Complex to implement

[EXAM TIP] *Know the three types of missing data: MCAR (completely random), MAR (depends on observed data), MNAR (depends on unobserved data). MNAR is the trickiest — the fact that data is missing IS informative.*

Encoding Categorical Variables

Method	When to Use	How It Works	Watch Out For
One-Hot Encoding	Nominal categories (no order), low cardinality	Creates binary column per category	Curse of dimensionality if >20 categories
Label / Ordinal Encoding	Ordinal categories (has order)	Maps to integers (0, 1, 2...)	Implies false ordering if data is nominal
Target Encoding	High cardinality categoricals	Replace category with mean of target	Data leakage — must use fold-based encoding
Frequency Encoding	High cardinality	Replace category with its count/frequency	Loses distinction between same-frequency categories
Binary Encoding	High cardinality	Convert to binary, then separate columns	Less interpretable

Feature Scaling

Method	Formula	Range	When to Use	Sensitive to Outliers?
Min-Max Normalization	(x - min) / (max - min)	[0, 1]	Neural networks, KNN, algorithms needing bounded input	Yes
Standardization (Z-score)	$(x - \mu) / \sigma$	Unbounded (mean=0, std=1)	Linear regression, logistic regression, SVM, PCA	Somewhat
Robust Scaling	(x - median) / IQR	Unbounded	Data with many outliers	No (uses median/IQR)

[EXAM TIP] *Tree-based algorithms (XGBoost, Random Forest) do NOT need feature scaling. Linear models, neural networks, KNN, and SVM DO need scaling. This is a very common exam question.*

Dimensionality Reduction

Technique	Type	How It Works	Use Case
PCA	Linear, unsupervised	Finds orthogonal axes of maximum variance	Reduce features before training; remove multicollinearity
t-SNE	Non-linear, unsupervised	Preserves local structure for visualization	2D/3D visualization of high-dimensional data ONLY
Feature Selection (filter)	Statistical	Correlation, mutual information, chi-squared	Remove irrelevant/redundant features
Feature Selection (wrapper)	Model-based	Forward/backward selection, RFE	Find optimal feature subset

[EXAM TIP] *PCA is for dimensionality reduction in TRAINING pipelines. t-SNE is for VISUALIZATION only — never use it as a preprocessing step for models. The exam tests this distinction.*

Handling Imbalanced Data

Technique	Category	How It Works	When to Use
SMOTE	Oversampling	Synthesizes new minority samples using KNN	Moderate imbalance (1:10 to 1:100)
Random Oversampling	Oversampling	Duplicates minority samples	Simple baseline, risk of overfitting

Technique	Category	How It Works	When to Use
Random Undersampling	Undersampling	Removes majority samples	Very large datasets where you can afford to lose data
Class Weights	Algorithm-level	Penalizes misclassification of minority class more	Most algorithms support this; no data manipulation needed
Anomaly Detection framing	Re-framing	Treat minority as anomaly (e.g., Random Cut Forest)	Extreme imbalance (1:10000+)
Ensemble methods	Algorithm-level	Balanced Random Forest, EasyEnsemble	When single model underperforms

[LIFE SCIENCES EXAMPLE] A hospital's readmission prediction dataset has these challenges — walk through each: (1) 30% of patients are missing 'smoking_status' — this is likely MNAR because smokers may avoid reporting. Create an indicator variable + impute with mode. (2) 'diagnosis_code' has 12,000 ICD-10 codes — too many for one-hot encoding. Group into top-level categories (circulatory, respiratory, etc.) then one-hot encode ~20 categories. (3) 'lab_values' like creatinine and glucose are on completely different scales — standardize for logistic regression, but note that if using XGBoost, scaling isn't needed.

[LIFE SCIENCES EXAMPLE] Drug discovery datasets are extremely imbalanced — only ~0.1% of candidate molecules are effective. Ask students: If accuracy is 99.9%, is the model good? (No — it's predicting 'inactive' for everything.) What metric should we use? (Recall to catch all active molecules, or F1 for balance.) What techniques help? (SMOTE to synthesize active molecule representations, class weights, or frame as anomaly detection.)

2.3 Domain 2 Quiz (15 min)

Question 1:

A dataset has a feature 'annual_income' that ranges from $20,000 to $5,000,000 with a strong right skew. Which transformation should be applied FIRST?

(A) Min-max normalization

(B) Standardization (z-score)

(C) Log transformation

(D) One-hot encoding

Correct Answer: (C)

Explanation: Log transformation reduces right skew and makes the distribution more normal. After log transform, then apply standardization or normalization. Min-max would still be skewed.

Question 2:

A feature 'hospital_department' has 500 unique values. What is the BEST encoding strategy?

(A) One-hot encoding all 500 values

(B) Label encoding

(C) Group into top-level categories, then one-hot encode

(D) Drop the feature entirely

Correct Answer: (C)

Explanation: 500 one-hot columns creates curse of dimensionality. Label encoding implies false ordering. Grouping into meaningful categories (e.g., surgical, medical, ICU) then one-hot encoding is the best approach.

Question 3:

Which dimensionality reduction technique should NEVER be used as a preprocessing step for model training?

(A) PCA

(B) t-SNE

(C) Feature selection

(D) Autoencoders

Correct Answer: (B)

Explanation: t-SNE is for visualization only — it doesn't preserve global structure, isn't deterministic, and can't transform new data points. PCA, feature selection, and autoencoders are all valid preprocessing steps.

Question 4:

A fraud detection model has 99.5% accuracy. There are 10,000 transactions: 50 fraudulent (positive) and 9,950 legitimate (negative). The model predicts ALL transactions as legitimate. What is the model's recall?

(A) 99.5%

(B) 0%

(C) 50%

(D) 100%

Correct Answer: (B)

Explanation: Recall = TP / (TP + FN). The model catches 0 fraud cases (TP = 0, FN = 50), so recall = 0/50 = 0%. This shows why accuracy is misleading for imbalanced datasets.

Question 5:

Which missing data mechanism is present when patients with more severe disease are less likely to complete follow-up surveys?

(A) Missing Completely at Random (MCAR)

(B) Missing at Random (MAR)

(C) Missing Not at Random (MNAR)

(D) This is not a missing data problem

Correct Answer: (C)

Explanation: MNAR: the probability of data being missing depends on the unobserved value itself (severity). Sicker patients drop out, so the missingness is related to the outcome. This is the hardest type to handle.

Question 6:

A data scientist is training a logistic regression model. Features include age (20-90), income ($20K-$500K), and number of visits (1-5). Which preprocessing step is MOST critical?

(A) One-hot encode all features

(B) Apply feature scaling (standardization)

(C) Apply PCA

(D) Apply SMOTE

Correct Answer: (B)

Explanation: Logistic regression is sensitive to feature scales. Income (20K-500K) would dominate age (20-90) and visits (1-5) without scaling. Standardization ensures equal contribution. One-hot is for categoricals. PCA and SMOTE are not indicated here.

Question 7:

A correlation heatmap shows that features A and B have a Pearson correlation of 0.97. What should the data scientist do?

(A) Remove both features

(B) Keep both features — more features are always better

(C) Remove one of the two features or use PCA

(D) Apply one-hot encoding to both

Correct Answer: (C)

Explanation: High correlation (0.97) indicates multicollinearity. Keeping both adds redundancy and can destabilize linear models. Remove one or use PCA to combine them. Never remove both — the information is still useful.

Domain 3: Modeling

Time Allocation: 90 minutes | Exam Weight: 36% | Questions: ~23 of 65

This is the largest and most important domain. It covers algorithm selection, SageMaker training, model evaluation, and deep learning.

3.1 Algorithm Selection (25 min)

[INSTRUCTOR NOTE] This section is critical. Students must be able to map a problem description to the correct algorithm. Use the tables as flashcard-style review.

Supervised Learning — Regression Algorithms

Algorithm	Type	Use Case	Key Feature	Exam Keyword
Linear Learner	Linear	Regression + classification	Built-in normalization, L1/L2 regularization	'linear relationship', 'simple baseline'
XGBoost	Tree ensemble	Structured/tabular data	Handles missing values, feature importance	'tabular data', 'structured', 'feature importance'
KNN	Instance-based	Small datasets, non-linear	No training phase; lazy learner	'similar items', 'recommendation'
DeepAR	RNN-based	Probabilistic time series forecasting	Handles multiple related time series	'forecast', 'time series', 'multiple series'
Random Cut Forest	Tree ensemble	Anomaly detection (unsupervised)	Assigns anomaly score	'anomaly', 'outlier detection'

Supervised Learning — Classification Algorithms

Algorithm	Type	Use Case	Key Feature	Exam Keyword
Linear Learner	Linear	Binary/multi-class classification	Fast, scalable, built-in regularization	'classify', 'linear boundary'
XGBoost	Tree ensemble	Tabular classification (most common)	Top Kaggle algorithm; handles mixed data	'tabular', 'structured features', 'best accuracy'
Image Classification	CNN (ResNet)	Image categorization	Transfer learning, fine-tuning	'classify images', 'image labels'
Object Detection	CNN (SSD/YOLO)	Localize + classify objects in images	Bounding boxes + labels	'locate objects', 'bounding boxes'
Semantic Segmentation	CNN (FCN)	Pixel-level image classification	Labels every pixel	'pixel-level', 'segmentation mask'

Algorithm	Type	Use Case	Key Feature	Exam Keyword
Factorization Machines	Matrix factorization	Sparse data, recommendations	High-dimensional sparse input	'sparse data', 'click prediction', 'recommendation'

Unsupervised Learning Algorithms

Algorithm	Type	Use Case	Key Feature	Exam Keyword
K-Means	Clustering	Group similar data points	Specify k clusters; uses web-scale k-means++	'cluster', 'group', 'segment'
PCA	Dim. reduction	Reduce features, remove noise	Projects to principal components	'reduce dimensions', 'too many features'
Random Cut Forest (RCF)	Anomaly detection	Detect outliers in data streams	Works on streaming data	'anomaly', 'outlier', 'fraud'
IP Insights	Anomaly detection	Detect anomalous IP usage patterns	Learns IP-entity associations	'suspicious IP', 'account takeover'
LDA (Latent Dirichlet Allocation)	Topic modeling	Discover topics in document corpus	Unsupervised topic discovery	'topics', 'themes', 'document clustering'

NLP & Sequence Algorithms

Algorithm	Type	Use Case	Mode	Exam Keyword
BlazingText	Word2Vec / Text Classification	Word embeddings OR text classification	Supervised (classify) or Unsupervised (embed)	'text classification', 'word embeddings', 'sentiment'
Seq2Seq	Encoder-Decoder RNN	Machine translation, summarization	Supervised	'translate', 'summarize', 'sequence to sequence'
Object2Vec	Embedding	Learn embeddings for arbitrary objects	Supervised	'embed pairs', 'similarity between entities'
Neural Topic Model (NTM)	Topic modeling (NN-based)	Discover topics (neural approach)	Unsupervised	'topics', 'themes' (alternative to LDA)

Algorithm Selection Decision Framework

Ask these questions in order:

1. What is the PROBLEM TYPE? → Classification, Regression, Clustering, Anomaly Detection, NLP, CV, Time Series, Recommendation

2. What is the DATA TYPE? → Tabular/structured, Images, Text, Time Series, Sparse

3. Is data LABELED? → Yes = Supervised, No = Unsupervised

4. How much DATA do you have? → Small = simpler models (Linear Learner, KNN), Large = complex (XGBoost, Deep Learning)

5. Do you need INTERPRETABILITY? → Yes = Linear Learner, XGBoost (feature importance), No = Deep Learning

Rapid-Fire Flashcards (Instructor Q&A)

[INSTRUCTOR NOTE] Read the left column, have students shout out the answer from the right column. Go fast!

Question	Answer
Tabular data, classification or regression?	XGBoost (default choice for structured data)
Forecast multiple related time series?	DeepAR
Detect anomalies in streaming data?	Random Cut Forest (RCF)
Classify images into categories?	Image Classification (ResNet)
Find objects + bounding boxes in images?	Object Detection (SSD)
Classify text documents (e.g., sentiment)?	BlazingText (supervised mode)
Generate word embeddings?	BlazingText (unsupervised mode / Word2Vec)
Discover topics in document corpus?	LDA or NTM
Translate text from one language to another?	Seq2Seq (or Amazon Translate for managed)
Cluster customers into segments?	K-Means

[LIFE SCIENCES EXAMPLE] Walk through algorithm selection for common life sciences problems: (1) Predict whether a patient will respond to chemotherapy based on genomic markers, demographics, and lab values → XGBoost (structured tabular data, classification). (2) Forecast ICU bed demand across 50 hospitals over the next 30 days → DeepAR (multiple related time series). (3) Cluster patients with similar disease progression patterns → K-Means. (4) Detect anomalous readings from medical devices → Random Cut Forest. (5) Classify pathology slide images as benign/malignant → Image Classification with transfer learning from ResNet.

[LIFE SCIENCES EXAMPLE] A pharmaceutical company wants to analyze 500,000 adverse event reports to discover common themes/topics. Which algorithm? (LDA or NTM for topic modeling.) They also want to classify new incoming reports by severity. (BlazingText for fast text classification.) Walk through why Seq2Seq is wrong here — it's for translation/summarization, not classification or topic discovery.

3.2 SageMaker Training & Tuning (20 min)

SageMaker Training Architecture

SageMaker training flow: S3 (training data) → Training Instance (Docker container with algorithm) → S3 (model artifacts). Key points:

- Training data is pulled FROM S3 to the training instance
- Model artifacts are written TO S3 after training
- Each training job runs in its own isolated container
- You choose the instance type, count, and input mode

Input Modes

Input Mode	How It Works	When to Use	Exam Keyword
File Mode	Downloads entire dataset to instance disk (EBS)	Small-medium datasets that fit on disk	'download data', default mode
Pipe Mode	Streams data directly from S3 (no disk copy)	Large datasets; RecordIO-Protobuf format	'stream from S3', 'large dataset', 'reduce startup'
FastFile Mode	POSIX file access backed by S3 streaming	Large datasets, any format (replaces Pipe)	'fast', 'large dataset', 'any format'

[EXAM TIP] *Pipe Mode streams data from S3, avoiding the need to download the entire dataset. It requires RecordIO-Protobuf format. FastFile Mode is newer and supports any format with similar performance benefits.*

Instance Types for ML

Instance Family	Optimized For	Use Case	Example
ml.m5.*	General purpose (balanced)	XGBoost, Linear Learner, preprocessing	ml.m5.xlarge
ml.c5.*	Compute-optimized (CPU)	CPU-intensive training (NLP tokenization)	ml.c5.2xlarge
ml.p3.* / ml.p4d.*	GPU (NVIDIA V100/A100)	Deep learning, CNNs, transformers	ml.p3.2xlarge
ml.g4dn.* / ml.g5.*	GPU (cost-effective)	Inference, lighter GPU training	ml.g4dn.xlarge
ml.inf1.* / ml.inf2.*	AWS Inferentia chips	Cost-optimized inference only	ml.inf1.xlarge
ml.trn1.*	AWS Trainium chips	Cost-optimized training	ml.trn1.2xlarge

[EXAM TIP] *GPUs (p3/p4d) for deep learning training. CPUs (m5/c5) for tree-based algorithms like XGBoost. Inferentia (inf1/inf2) for cost-optimized inference. The exam frequently tests instance selection.*

Distributed Training

Strategy	What Gets Split	When to Use	SageMaker Support
Data Parallelism	Data split across GPUs, model replicated	Model fits in 1 GPU, but data is huge	SageMaker Distributed Data Parallel Library
Model Parallelism	Model split across GPUs	Model too large for 1 GPU (LLMs, huge CNNs)	SageMaker Distributed Model Parallel Library

Automatic Model Tuning (Hyperparameter Optimization)

Strategy	How It Works	When to Use
Bayesian Optimization	Builds probabilistic model of objective function; smart exploration	Default choice; best for <500 total jobs
Random Search	Randomly samples hyperparameter space	Good starting point; embarrassingly parallel
Hyperband	Early stopping of bad trials + resource allocation	Large search space; time-constrained
Grid Search	Exhaustive search of all combinations	Small search space (<20 combinations)

[EXAM TIP] *SageMaker's Automatic Model Tuning uses Bayesian optimization by default. It's the most sample-efficient strategy (finds good parameters in fewer trials). Know that it works by treating HPO as a regression problem.*

Key Hyperparameters by Algorithm

Algorithm	Key Hyperparameters	What They Control
XGBoost	max_depth, eta (learning rate), num_round, subsample, min_child_weight	Tree depth, learning speed, iterations, regularization
Linear Learner	learning_rate, l1, wd (L2), mini_batch_size, epochs	Optimization speed, regularization
K-Means	k, init_method, epochs	Number of clusters, initialization strategy

Algorithm	Key Hyperparameters	What They Control
Image Classification	num_classes, num_training_samples, epochs, learning_rate, optimizer	Architecture, training config
DeepAR	context_length, prediction_length, epochs, num_cells	Input window, forecast horizon, network size
BlazingText	mode (supervised/skipgram/cbow), vector_dim, epochs, learning_rate	Algorithm mode, embedding size

[LIFE SCIENCES EXAMPLE] Training a deep learning model for medical image analysis (chest X-ray classification) on 2 million images: (1) Instance type? → ml.p3.2xlarge or ml.p4d for GPU (CNNs need GPUs). (2) Input mode? → Pipe Mode to avoid downloading 2M images to disk. (3) Cost optimization? → Managed Spot Training with checkpointing to S3. (4) The model fits in one GPU but data is huge → Data Parallelism (split images across GPUs, synchronize gradients). Walk through the cost savings: a p3.2xlarge at ~$3.83/hr vs Spot at ~$1.15/hr — training a 24-hour job saves ~$64.

3.3 Model Evaluation & Regularization (20 min)

Classification Metrics

Metric	Formula	When to Optimize	Life Sciences Example
Accuracy	Correct / Total	Balanced classes only	Not useful for rare disease detection
Precision	TP / (TP + FP)	Cost of false positive is high	Drug safety: don't approve unsafe drugs
Recall (Sensitivity)	TP / (TP + FN)	Cost of false negative is high	Cancer screening: don't miss cancer
F1 Score	2 * (P*R) / (P+R)	Balance precision and recall	General clinical prediction
AUC-ROC	Area under ROC curve	Overall model discrimination	Comparing model versions
Specificity	TN / (TN + FP)	True negative rate matters	Healthy patient confirmation

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN) — Type II Error

	Predicted Positive	Predicted Negative
Actual Negative	False Positive (FP) — Type I Error	True Negative (TN)

[EXAM TIP] *Memorize: Precision = TP/(TP+FP) — 'Of those I called positive, how many were right?' Recall = TP/(TP+FN) — 'Of all actual positives, how many did I find?' The exam WILL ask you to choose the right metric for a scenario.*

Regression Metrics

Metric	Formula Concept	Interpretation	Sensitivity to Outliers
MSE (Mean Squared Error)	Average of squared errors	Lower is better; penalizes large errors	Very sensitive
RMSE (Root MSE)	Square root of MSE	Same units as target; most common	Sensitive
MAE (Mean Absolute Error)	Average of absolute errors	Robust to outliers	Less sensitive
R^2 (Coefficient of Determination)	1 - (SS_res / SS_tot)	% of variance explained; 1.0 is perfect	Moderate
MAPE	Average of |error/actual| %	Percentage error; interpretable	Undefined if actual = 0

Overfitting vs. Underfitting

Symptom	Overfitting	Underfitting
Training performance	Very high (near perfect)	Low
Validation performance	Significantly lower than training	Low (similar to training)
Gap	Large train-val gap	Small gap, both bad
Model complexity	Too complex (memorizing noise)	Too simple (can't capture patterns)
Bias-Variance	Low bias, high variance	High bias, low variance
Solutions	Regularization, more data, simpler model, dropout, early stopping	More features, more complex model, more training time, less regularization

Regularization Techniques

Technique	How It Works	Effect	Best For
L1 (Lasso)	Adds |weights| penalty to loss	Drives weights to zero (feature selection)	When many irrelevant features exist
L2 (Ridge)	Adds weights² penalty to loss	Shrinks weights toward zero (never exactly zero)	When all features are somewhat useful
Elastic Net	Combines L1 + L2	Both feature selection and shrinkage	When unsure; best of both worlds
Dropout	Randomly zeros neurons during training	Prevents co-adaptation of neurons	Neural networks
Early Stopping	Stop training when validation loss stops improving	Prevents overtraining	Any iterative algorithm
Data Augmentation	Create modified copies of training data	Effectively increases training set size	Image classification (rotate, flip, crop)

[EXAM TIP] *L1 = Lasso = feature selection (zeroes out weights). L2 = Ridge = weight shrinkage. Remember: L1 has 1 sharp corner (sparse), L2 is smooth (dense). Elastic Net = L1 + L2 combined.*

Validation Strategies

Strategy	How It Works	When to Use
Train/Validation/Test Split	70/15/15 or 80/10/10 random split	Large datasets (>100K samples)
K-Fold Cross-Validation	Rotate through k folds, train on k-1	Medium datasets; more robust estimate
Stratified K-Fold	K-fold preserving class distribution	Imbalanced classification
Time Series Split	Train on past, validate on future (no shuffle)	Time series data (prevent data leakage)

[EXAM TIP] *For time series, NEVER shuffle data. Always train on earlier data, validate on later data. Shuffling causes data leakage because the model sees future values during training.*

[LIFE SCIENCES EXAMPLE] Cancer screening model — which metric matters most? Walk through the consequence matrix: False Negative = missed cancer = patient doesn't get treatment = potentially fatal → must maximize RECALL. False Positive = unnecessary biopsy = stress and cost but not fatal → tolerable. So optimize for recall, accept lower precision. Contrast with: spam filter for a healthcare provider's email — FP = important patient message goes to spam = dangerous → optimize PRECISION.

[LIFE SCIENCES EXAMPLE] A drug interaction prediction model has 200 molecular features but only 500 training samples. It's overfitting (98% training accuracy, 61% validation). Solutions: (1) L1 regularization to zero out irrelevant molecular features (feature selection). (2) PCA to reduce 200 features to 20 principal components. (3) Dropout if using a neural network. (4) Transfer learning — use a model pre-trained on a large chemistry dataset. Ask students: which solution addresses the root cause (too many features, too few samples) most directly? (L1 + getting more data.)

3.4 Deep Learning Concepts (15 min)

Activation Functions

Function	Range	Use Case	Pros	Cons
ReLU	[0, ∞)	Default for hidden layers	Fast, avoids vanishing gradient	Dead neurons (output 0 permanently)
Leaky ReLU	(-∞, ∞)	Alternative to ReLU	No dead neurons	Slightly more complex
Sigmoid	(0, 1)	Binary classification output	Probability output	Vanishing gradient, slow
Tanh	(-1, 1)	Hidden layers (legacy)	Zero-centered	Vanishing gradient
Softmax	(0, 1) per class, sum=1	Multi-class classification output	Probability distribution over classes	Only for output layer

[EXAM TIP] *ReLU for hidden layers (default). Sigmoid for binary output. Softmax for multi-class output. The exam tests this mapping.*

CNN Architecture (Convolutional Neural Networks)

CNN pipeline: Input Image → [Convolution → ReLU → Pooling] × N → Flatten → Fully Connected → Output

Layer	Purpose	Key Parameter
Convolutional	Extract local features (edges, textures, patterns)	Number of filters, kernel size
Pooling (Max/Avg)	Reduce spatial dimensions, add translation invariance	Pool size (typically 2×2)
Flatten	Convert 2D feature maps to 1D vector	N/A
Fully Connected (Dense)	Learn non-linear combinations for classification	Number of neurons

Layer	Purpose	Key Parameter
Output	Final prediction	Sigmoid (binary) or Softmax (multi-class)

RNN / LSTM / GRU

Architecture	Handles	Key Innovation	Use Case
RNN (Vanilla)	Sequences	Recurrent connections	Simple sequences (rarely used alone)
LSTM	Long sequences	Cell state + 3 gates (forget, input, output)	NLP, time series, speech (long dependencies)
GRU	Long sequences	2 gates (reset, update) — simpler than LSTM	Same as LSTM but faster to train

[EXAM TIP] *LSTM and GRU solve the vanishing gradient problem for sequences. LSTM has 3 gates, GRU has 2. GRU is faster but LSTM is more expressive. For the exam, know that both handle long-range dependencies.*

Transfer Learning

Use a model pre-trained on a large dataset (e.g., ImageNet) and fine-tune on your smaller, domain-specific dataset.

- Freeze early layers (generic features like edges, textures)
- Replace and train the final classification layer(s) for your task
- Optionally unfreeze and fine-tune later layers with a low learning rate
- Dramatically reduces training time and data requirements

[EXAM TIP] *SageMaker Image Classification and Object Detection support transfer learning by default. Always use it when you have limited labeled data.*

Optimizers

Optimizer	Key Feature	When to Use
SGD	Simple gradient descent with optional momentum	Baseline; good with learning rate schedules
Adam	Adaptive learning rates per parameter	Default choice for most deep learning
RMSprop	Adaptive learning rates (predecessor to Adam)	RNNs (sometimes preferred over Adam)

Optimizer	Key Feature	When to Use
Adagrad	Adapts learning rate based on feature frequency	Sparse data (NLP embeddings)

Vanishing & Exploding Gradients

Problem	Symptom	Cause	Solutions
Vanishing Gradients	Early layers stop learning; loss plateaus	Repeated multiplication by small values in deep networks	ReLU activation, LSTM/GRU, residual connections (ResNet), batch normalization
Exploding Gradients	Loss becomes NaN; weights oscillate wildly	Repeated multiplication by large values	Gradient clipping, lower learning rate, batch normalization

[LIFE SCIENCES EXAMPLE] Medical imaging is the #1 deep learning application in life sciences. Walk through a pathology pipeline: (1) Input: microscopy slide images (very high resolution, often 100,000×100,000 pixels — need to tile into patches). (2) Architecture: CNN (ResNet via SageMaker Image Classification). (3) Transfer learning: freeze ResNet layers pre-trained on ImageNet, fine-tune final layers on pathology images. (4) Output activation: sigmoid for binary (benign/malignant), softmax for multi-class (normal/benign/in-situ/invasive). (5) Key challenge: vanishing gradients in deep networks → ResNet's skip connections solve this.

3.5 Domain 3 Quiz (15 min)

Question 1:

A company has a tabular dataset with 50 numerical features and wants to predict customer churn (binary classification). Which SageMaker algorithm is the BEST first choice?

(A) Linear Learner

(B) XGBoost

(C) Image Classification

(D) BlazingText

Correct Answer: (B)

Explanation: XGBoost is the go-to algorithm for structured/tabular data. It handles mixed feature types, provides feature importance, and typically achieves the best accuracy on tabular datasets.

Question 2:

A hospital needs to forecast patient admissions across 200 locations for the next 14 days. Historical data spans 3 years. Which algorithm should they use?

(A) Linear Learner

(B) K-Means

(C) DeepAR

(D) XGBoost

Correct Answer: (C)

Explanation: DeepAR is designed for forecasting multiple related time series. It learns across all 200 locations, capturing shared patterns. XGBoost could work but doesn't natively handle the temporal + multi-series nature.

Question 3:

Which SageMaker training input mode streams data directly from S3 without downloading it to the training instance?

(A) File Mode

(B) Pipe Mode

(C) S3 Direct Mode

(D) EBS Mode

Correct Answer: (B)

Explanation: Pipe Mode streams data from S3 directly, reducing startup time and disk requirements. File Mode downloads everything first. FastFile Mode also streams but uses POSIX access.

Question 4:

A model achieves 95% training accuracy but only 60% validation accuracy. What is this called and what should you try FIRST?

(A) Underfitting — add more features

(B) Overfitting — add regularization (L1/L2, dropout)

(C) Overfitting — increase model complexity

(D) Underfitting — train longer

Correct Answer: (B)

Explanation: Large gap between training (95%) and validation (60%) = overfitting. The model memorizes training data. Regularization (L1, L2, dropout) constrains model complexity. Adding complexity would make it worse.

Question 5:

For a cancer screening model, which metric should be MAXIMIZED?

(A) Accuracy

(B) Precision

(C) Recall (Sensitivity)

(D) Specificity

Correct Answer: (C)

Explanation: In cancer screening, missing a true cancer case (False Negative) is far worse than a false alarm (False Positive). Recall = TP/(TP+FN) measures how many actual positives are caught.

Question 6:

Which regularization technique performs FEATURE SELECTION by driving irrelevant feature weights to exactly zero?

(A) L2 (Ridge)

(B) L1 (Lasso)

(C) Dropout

(D) Early Stopping

Correct Answer: (B)

Explanation: L1 (Lasso) regularization adds |w| penalty, which has a sharp corner at zero, causing some weights to become exactly 0. L2 shrinks weights toward zero but never reaches exactly 0.

Question 7:

What activation function should be used in the OUTPUT layer of a multi-class classification neural network?

(A) ReLU

(B) Sigmoid

(C) Softmax

(D) Tanh

Correct Answer: (C)

Explanation: Softmax outputs a probability distribution across all classes (sum = 1). Sigmoid is for binary classification. ReLU and Tanh are for hidden layers, not output.

Question 8:

SageMaker Automatic Model Tuning uses which optimization strategy by default?

(A) Grid Search

(B) Random Search

(C) Bayesian Optimization

(D) Evolutionary Algorithm

Correct Answer: (C)

Explanation: Bayesian optimization builds a surrogate model of the objective function, making it more sample-efficient than random or grid search. It finds good hyperparameters in fewer trials.

Question 9:

A deep learning model for pathology images uses ResNet. What problem do ResNet's skip connections (residual connections) solve?

(A) Overfitting

(B) Vanishing gradients

(C) Data imbalance

(D) Feature scaling

Correct Answer: (B)

Explanation: Skip connections allow gradients to flow directly through shortcut paths, preventing them from vanishing in very deep networks. This enables training of networks with 50-150+ layers.

Question 10:

Which distributed training strategy should be used when the MODEL is too large to fit on a single GPU?

(A) Data Parallelism

(B) Model Parallelism

(C) Pipe Mode

(D) Multi-Model Endpoints

Correct Answer: (B)

Explanation: Model Parallelism splits the model across multiple GPUs. Data Parallelism replicates the model and splits the data. Pipe Mode is about data input, not distributed training.

BREAK — 10 minutes

> **[INSTRUCTOR NOTE]** Students may be fatigued after the dense modeling section. Encourage them to stand, stretch, and review their notes. You should be at approximately 3:20 into the session.

Domain 4: ML Implementation & Operations

Time Allocation: 55 minutes | Exam Weight: 20% | Questions: ~13 of 65

4.1 Model Deployment (20 min)

Inference Options

Option	Latency	Use Case	Scaling	Exam Keyword
Real-Time Endpoint	Milliseconds	Live applications, APIs	Auto-scaling (min/max instances)	'low latency', 'real-time', 'API'

Option	Latency	Use Case	Scaling	Exam Keyword
Serverless Inference	Cold start possible	Infrequent/unpredictable traffic	Scales to zero (pay per request)	'intermittent traffic', 'cost-sensitive'
Batch Transform	Minutes–hours	Process large datasets offline	Managed cluster	'batch predictions', 'score entire dataset'
Async Inference	Seconds–minutes	Large payloads, queued processing	Queue-based, auto-scaling	'large payload', 'queue', 'near-real-time'

[EXAM TIP] *Real-time = always running, lowest latency. Batch = offline bulk scoring. Serverless = intermittent traffic, cold starts OK. Async = large payloads or long inference time (e.g., video processing).*

Deployment Strategies

Strategy	How It Works	Risk Level	Use Case
Blue/Green	Two full environments; switch traffic atomically	Low (instant rollback)	Production deployments with zero downtime
A/B Testing	Split traffic between model versions by percentage	Medium	Comparing model performance on live traffic
Canary	Route small % to new version, gradually increase	Low	Gradual rollout with monitoring
Shadow (Dark Launch)	Send traffic to both; only serve old model's results	Very Low	Testing new model without impacting users

Multi-Model Endpoints

Host multiple models on a single endpoint. Models are loaded/unloaded dynamically from S3. Key benefits:

- Cost savings: share infrastructure across hundreds of models
- Models loaded on-demand; LRU cache eviction for less-used models
- All models must use the same framework (e.g., all XGBoost or all PyTorch)
- Latency: first request to an unloaded model has cold start

Model Optimization for Deployment

Technique	What It Does	Benefit	SageMaker Service
SageMaker Neo	Compiles model for specific hardware (x86, ARM, GPU)	2–25x faster inference, smaller model	SageMaker Neo
Quantization	Reduces numerical precision (FP32 → FP16 or INT8)	Smaller model, faster inference	Built into Neo / manual
Pruning	Removes near-zero weights from neural network	Smaller, faster model with minimal accuracy loss	Manual (framework-level)
Knowledge Distillation	Train small 'student' model to mimic large 'teacher'	Much smaller model with similar performance	Manual process

Edge Deployment

Service	Purpose	Use Case
SageMaker Edge Manager	Deploy, manage, and monitor models on edge devices	Fleet of edge devices running ML
AWS IoT Greengrass	Run Lambda + ML models on local devices	Offline inference at edge locations
AWS Panorama	Computer vision at the edge	Camera-based visual inspection, retail analytics

[LIFE SCIENCES EXAMPLE] A hospital deploys a sepsis early warning model: (1) Deployment type? → Real-time endpoint — sepsis prediction must be instant as new vitals arrive, not batch. (2) A/B testing: deploy the new model v2 alongside v1, route 10% of predictions to v2, compare alert accuracy for 2 weeks before full cutover. (3) Edge scenario: a rural clinic has unreliable internet. Deploy the model to an IoT Greengrass device at the clinic using SageMaker Neo to compile for ARM. The model runs locally even when disconnected.

[LIFE SCIENCES EXAMPLE] A lab has 200 different protein-ligand binding models, one per target protein. Deploying 200 endpoints is expensive. Solution: Multi-Model Endpoint — host all 200 models on a single endpoint, load/unload dynamically based on which protein is being analyzed. Cost savings: ~95% vs separate endpoints.

4.2 MLOps & Security (20 min)

SageMaker MLOps Services

Service	Purpose	Key Feature
SageMaker Pipelines	ML workflow orchestration (CI/CD for ML)	DAG-based, integrates with all SageMaker steps
Model Registry	Version, track, and approve models	Model approval workflow; stage: dev/staging/prod
SageMaker Experiments	Track training runs, parameters, metrics	Compare experiments side-by-side
SageMaker Projects	MLOps templates with CI/CD built-in	Pre-built templates for common patterns

Model Monitoring

Monitor Type	What It Detects	How It Works
Data Quality	Data drift (input distribution changes)	Compares live data statistics to training baseline
Model Quality	Model performance degradation	Compares predictions to ground truth labels
Bias Drift	Fairness metrics changing over time	Monitors Clarify bias metrics continuously
Feature Attribution Drift	Feature importance changing	Monitors SHAP values over time

[EXAM TIP] *Data drift = input distribution changes (e.g., new patient demographics). Model drift = accuracy drops. Bias drift = fairness metrics degrade. Know all four monitoring types.*

SageMaker Clarify

Detects bias and explains model predictions:

Capability	When	What It Does
Pre-training Bias Detection	Before training	Detects imbalances in training data across sensitive groups
Post-training Bias Detection	After training	Detects biased predictions (disparate impact, etc.)
SHAP Explanations	After training	Explains individual predictions via feature attribution
Continuous Monitoring	In production	Monitors bias and explainability metrics over time

Security for ML

Security Layer	Service/Feature	Implementation
Encryption at Rest	AWS KMS	S3 (SSE-S3/SSE-KMS), EBS volumes, model artifacts
Encryption in Transit	TLS 1.2+	All SageMaker API calls and data transfers
Network Isolation	VPC, Private Subnets	Training/inference in VPC; S3 VPC endpoint (no internet)
Access Control	IAM Roles	SageMaker execution role; least-privilege per user
Network Isolation Mode	SageMaker setting	Prevents containers from making outbound network calls
Audit	CloudTrail	Log all SageMaker API calls

SageMaker Feature Store

Centralized repository for storing, sharing, and managing ML features:

- Online Store: Low-latency reads for real-time inference (DynamoDB-backed)
- Offline Store: Historical features for training (S3-backed, Glue Catalog)
- Feature Groups: Logical grouping of related features
- Ensures consistency between training and inference features

AWS AI Services (Managed ML)

Service	What It Does	Exam Keyword
Amazon Rekognition	Image/video analysis (faces, objects, text, content moderation)	'analyze images', 'detect faces', 'content moderation'
Amazon Comprehend	NLP: sentiment, entities, key phrases, language detection	'sentiment analysis', 'entity extraction', 'NLP'
Comprehend Medical	Extract medical entities from clinical text (HIPAA eligible)	'clinical notes', 'medical NLP', 'PHI detection'
Amazon Translate	Neural machine translation	'translate text', 'multi-language'
Amazon Transcribe	Speech-to-text	'transcribe audio', 'speech to text'
Amazon Polly	Text-to-speech (neural voices)	'text to speech', 'generate audio'
Amazon Lex	Conversational chatbot (Alexa technology)	'chatbot', 'voice assistant'
Amazon Textract	OCR + structured data from documents	'extract text from documents', 'forms', 'tables'

Service	What It Does	Exam Keyword
Amazon Forecast	Time series forecasting (managed)	'forecast', 'no ML expertise needed'
Amazon Personalize	Recommendations (managed)	'recommendations', 'personalization'
Amazon Fraud Detector	Fraud detection (managed)	'detect fraud', 'online payments'
Amazon Kendra	Intelligent enterprise search	'enterprise search', 'document search'

[EXAM TIP] *If the question says 'no ML expertise' or 'quickly deploy' + a specific task (forecast, recommend, detect fraud, etc.), the answer is likely one of the managed AI services, NOT SageMaker.*

[LIFE SCIENCES EXAMPLE] HIPAA compliance for a healthcare ML system: (1) Encryption at rest → KMS-managed keys for S3 training data, EBS volumes on training instances, and model artifacts. (2) Encryption in transit → TLS enforced by SageMaker. (3) Network → VPC with private subnets, S3 VPC endpoint (no internet), network isolation mode on training jobs. (4) Access → SageMaker execution role with least-privilege IAM, separate roles per data scientist. (5) Monitoring → SageMaker Model Monitor to detect data drift (e.g., patient demographics shifting post-deployment, which could signal bias).

[LIFE SCIENCES EXAMPLE] A clinical AI model that predicts patient risk scores shows different false negative rates across racial groups. SageMaker Clarify detects this via Disparate Impact and Difference in Conditional Acceptance metrics. Walk through the remediation: (1) Pre-training: check for class imbalance across demographic groups in training data, augment underrepresented groups. (2) Post-training: adjust decision thresholds per group or retrain with balanced data. (3) Ongoing: deploy Clarify continuous monitoring to catch drift in bias metrics.

[LIFE SCIENCES EXAMPLE] Comprehend Medical is Amazon Comprehend's HIPAA-eligible variant that extracts medical entities (medications, conditions, procedures, dosages) from clinical text. Example: process physician notes to extract structured data → use as features for a readmission model. Also mention Amazon HealthLake (FHIR-based health data lake) as a storage option for healthcare data.

4.3 Domain 4 Quiz (15 min)

Question 1:

A hospital needs real-time predictions from a sepsis detection model with sub-second latency. Which SageMaker inference option is MOST appropriate?

(A) Batch Transform

(B) Real-Time Endpoint

(C) Serverless Inference

(D) Async Inference

Correct Answer: (B)

Explanation: Real-time endpoints provide millisecond latency for live applications. Batch is offline. Serverless has cold starts. Async is for large payloads, not sub-second latency.

Question 2:

A company wants to deploy a new model version but needs to test it on live traffic without risking patient safety. Which deployment strategy should they use?

(A) Blue/Green

(B) A/B Testing

(C) Shadow (Dark Launch)

(D) Direct replacement

Correct Answer: (C)

Explanation: Shadow deployment sends traffic to both models but only serves the old model's results. The new model's predictions are logged for comparison but never shown to users. Zero risk.

Question 3:

Which SageMaker feature detects when the input data distribution in production differs from the training data distribution?

(A) SageMaker Clarify

(B) SageMaker Model Monitor (Data Quality)

(C) SageMaker Debugger

(D) SageMaker Experiments

Correct Answer: (B)

Explanation: Model Monitor's Data Quality monitoring compares live input data statistics against a training baseline to detect data drift. Clarify is for bias/explainability. Debugger is for training issues.

Question 4:

An ML team needs to ensure their SageMaker training job cannot make any outbound network calls (for security compliance). Which feature should they enable?

(A) VPC mode

(B) Network Isolation mode

(C) IAM role restrictions

(D) Security groups

Correct Answer: (B)

Explanation: Network Isolation mode prevents the training container from making ANY outbound network calls. VPC restricts network to your VPC but still allows outbound within it. Security groups control port access but not outbound calls from the container.

Question 5:

A research lab has 300 different ML models (one per experiment). Deploying 300 separate endpoints is too expensive. What should they use?

(A) Batch Transform

(B) Multi-Model Endpoints

(C) Serverless Inference

(D) Elastic Inference

Correct Answer: (B)

Explanation: Multi-Model Endpoints host many models on a single endpoint, loading/unloading dynamically. This is far more cost-effective than 300 separate endpoints. Serverless could work but each model needs its own endpoint.

Question 6:

Which AWS AI service should be used to extract medication names, dosages, and conditions from unstructured physician notes?

(A) Amazon Comprehend

(B) Amazon Comprehend Medical

(C) Amazon Textract

(D) Amazon Rekognition

Correct Answer: (B)

Explanation: Comprehend Medical is specifically designed to extract medical entities (medications, conditions, dosages, procedures) from clinical text. Regular Comprehend handles general NLP. Textract extracts text from images/PDFs.

Question 7:

SageMaker Clarify detects that a clinical risk model has disparate false negative rates across demographic groups. Which metric is this measuring?

(A) Data Quality drift

(B) Feature Attribution drift

(C) Post-training bias (Disparate Impact)

(D) Model Quality drift

Correct Answer: (C)

Explanation: Different false negative rates across demographic groups is a post-training bias metric (specifically, Difference in Conditional Acceptance or Disparate Impact). Data Quality drift is about input distribution changes.

Question 8:

Which SageMaker service compiles models for specific target hardware (e.g., ARM processors) to optimize inference performance?

(A) SageMaker Debugger

(B) SageMaker Neo

(C) SageMaker Autopilot

(D) SageMaker Processing

Correct Answer: (B)

Explanation: SageMaker Neo compiles models for specific hardware targets (x86, ARM, GPU, edge devices), enabling 2–25x performance improvement. It's used for both cloud and edge deployment optimization.

Final Review

Time Allocation: 55 minutes

5.1 Service Cheat Sheet (10 min)

[INSTRUCTOR NOTE] Walk through this table quickly as a rapid-fire review. Students should use this as their primary study reference.

SageMaker Built-in Algorithms Quick Reference

#	Algorithm	Problem Type	Data Type	Key Exam Trigger
1	Linear Learner	Classification / Regression	Tabular	'linear', 'simple baseline'
2	XGBoost	Classification / Regression	Tabular	'structured data', 'tabular', 'best accuracy'
3	K-Nearest Neighbors	Classification / Regression	Tabular	'similar items', 'instance-based'
4	Factorization Machines	Classification / Regression	Sparse tabular	'sparse', 'recommendations', 'click prediction'
5	Image Classification	Classification	Images	'classify images', 'categorize photos'
6	Object Detection	Detection	Images	'bounding boxes', 'locate objects'

#	Algorithm	Problem Type	Data Type	Key Exam Trigger
7	Semantic Segmentation	Segmentation	Images	'pixel-level', 'segment regions'
8	DeepAR	Forecasting	Time Series	'forecast', 'multiple time series'
9	BlazingText	Classification / Embeddings	Text	'text classification', 'word embeddings'
10	Seq2Seq	Sequence-to-Sequence	Text	'translate', 'summarize'
11	Object2Vec	Embeddings	Pairs of objects	'similarity', 'embed pairs'
12	Neural Topic Model (NTM)	Topic Modeling	Text	'discover topics', 'themes'
13	LDA	Topic Modeling	Text	'topics' (alternative to NTM)
14	K-Means	Clustering	Any	'cluster', 'segment', 'group'
15	PCA	Dim. Reduction	Any	'reduce dimensions', 'too many features'
16	Random Cut Forest	Anomaly Detection	Any	'anomaly', 'outlier', 'unusual'
17	IP Insights	Anomaly Detection	IP addresses	'suspicious IP', 'login anomaly'

Key Distinctions to Remember

Comparison	Service A	Service B	Key Difference
Stream ingestion	Kinesis Data Streams	Kinesis Firehose	KDS = custom consumers + replay. Firehose = managed delivery to S3/Redshift
ETL	AWS Glue	Amazon EMR	Glue = serverless. EMR = managed cluster for heavy Spark/Hadoop
Text classification vs topics	BlazingText	LDA / NTM	BlazingText = classify. LDA/NTM = discover topics

Comparison	Service A	Service B	Key Difference
Image tasks	Image Classification	Object Detection	Classification = what is it? Detection = where is it? + what?
Regularization	L1 (Lasso)	L2 (Ridge)	L1 = feature selection (zeros). L2 = weight shrinkage (never zero)
Distributed training	Data Parallelism	Model Parallelism	Data = split data, replicate model. Model = split model across GPUs

5.2 Practice Exam (25 min)

> **[INSTRUCTOR NOTE]** This is a timed mini-exam. Give students 20 minutes to answer all 20 questions independently, then spend 5 minutes reviewing the hardest ones. Tell them: 1 minute per question, just like the real exam.

Question 1:

A company collects IoT sensor data from manufacturing equipment. They need real-time anomaly detection on the data stream AND batch storage in S3 for monthly reporting. Which architecture is MOST appropriate?

(A) IoT Core → Kinesis Data Streams → Lambda (RCF anomaly detection) + Kinesis Firehose → S3

(B) IoT Core → S3 → Lambda (scheduled anomaly detection)

(C) IoT Core → SQS → EC2 fleet → S3

(D) IoT Core → DynamoDB → DynamoDB Streams → Lambda

Correct Answer: (A)

Explanation: KDS handles the real-time stream for anomaly detection via Lambda. Firehose delivers to S3 for batch storage. Option B isn't real-time. Options C/D are over-engineered.

Question 2:

A data scientist has a CSV dataset with 200 columns and 10 million rows. They need to train an XGBoost model on SageMaker. What file format and input mode combination will give the FASTEST training?

(A) CSV with File Mode

(B) Parquet with File Mode

(C) RecordIO-Protobuf with Pipe Mode

(D) JSON with FastFile Mode

Correct Answer: (C)

Explanation: RecordIO-Protobuf is SageMaker's optimized format and supports Pipe Mode, which streams data from S3 without downloading. This combination minimizes both I/O and startup time.

Question 3:

A genomics dataset has 50,000 gene expression features but only 200 patient samples. The model overfits severely. Which approach BEST addresses the root cause?

(A) Increase the number of training epochs

(B) Use a more complex model (deeper neural network)

(C) Apply PCA to reduce to 50 components, and add L1 regularization

(D) Switch from XGBoost to Linear Learner

Correct Answer: (C)

Explanation: Root cause: far more features (50K) than samples (200). PCA reduces dimensionality drastically. L1 regularization further selects relevant features. More epochs or complexity would worsen overfitting.

Question 4:

A pharmaceutical company wants to classify 1 million adverse event reports by severity (low/medium/high/critical). The text is short (1-3 sentences). Which SageMaker algorithm is BEST?

(A) Seq2Seq

(B) BlazingText (supervised mode)

(C) LDA

(D) Object2Vec

Correct Answer: (B)

Explanation: BlazingText supervised mode is designed for fast text classification. Seq2Seq is for translation/generation. LDA is for topic discovery (unsupervised). Object2Vec is for embedding pairs.

Question 5:

A hospital's readmission prediction model shows 92% accuracy but only 35% recall for readmitted patients. The readmission rate is 12%. What is the MOST LIKELY issue?

(A) The model is underfitting

(B) The model is overfitting to the majority class (non-readmitted)

(C) The feature engineering is incorrect

(D) The model needs more training epochs

Correct Answer: (B)

Explanation: With 88% non-readmitted patients, achieving 92% accuracy is easy by mostly predicting 'not readmitted.' Low recall (35%) for the minority class confirms the model is biased toward the majority class. Solutions: class weights, SMOTE, threshold adjustment.

Question 6:

Which SageMaker feature should be used to track and compare multiple training runs with different hyperparameters?

(A) SageMaker Pipelines

(B) SageMaker Experiments

(C) SageMaker Model Registry

(D) SageMaker Debugger

Correct Answer: (B)

Explanation: SageMaker Experiments is designed to organize, track, and compare training runs. Pipelines orchestrate workflows. Model Registry manages model versions. Debugger identifies training issues.

Question 7:

A data scientist notices that a feature 'patient_weight' has values ranging from 50-120 (kg) while 'white_blood_cell_count' ranges from 4,000-11,000. They plan to use KNN. What preprocessing step is CRITICAL?

(A) Log transformation

(B) One-hot encoding

(C) Feature scaling (standardization or normalization)

(D) PCA

Correct Answer: (C)

Explanation: KNN uses distance calculations. Without scaling, WBC count (4000-11000) would dominate the distance calculation over weight (50-120). Standardization puts both on comparable scales.

Question 8:

An ML team needs to deploy a model that processes medical images (each ~50MB). Predictions take ~30 seconds. Traffic is sporadic. Which inference option is BEST?

(A) Real-Time Endpoint

(B) Serverless Inference

(C) Batch Transform

(D) Async Inference

Correct Answer: (D)

Explanation: Async Inference handles large payloads and long processing times. It queues requests and processes them. Real-time endpoints have a 60s/6MB default limit. Serverless has similar limits. Batch is for offline processing, not sporadic live requests.

Question 9:

Which of the following CORRECTLY describes the difference between L1 and L2 regularization?

(A) L1 shrinks all weights equally; L2 drives some weights to zero

(B) L1 drives some weights to exactly zero (feature selection); L2 shrinks weights toward zero but never exactly zero

(C) L1 is used for neural networks; L2 is used for linear models

(D) L1 and L2 are functionally identical

Correct Answer: (B)

Explanation: L1 (Lasso) adds $|w|$ penalty with a sharp corner at zero, enabling automatic feature selection. L2 (Ridge) adds w^2 penalty which is smooth, shrinking weights but never reaching exactly zero.

Question 10:

A model deployed in production starts making significantly worse predictions after 3 months, even though the model hasn't changed. What is the MOST LIKELY cause?

(A) The model's weights have decayed

(B) The training data was corrupted

(C) Data drift — the input data distribution has changed

(D) The endpoint instance is underpowered

Correct Answer: (C)

Explanation: Data drift is the most common cause of production model degradation. The real-world data has shifted from what the model was trained on. SageMaker Model Monitor (Data Quality) detects this.

Question 11:

A deep learning model for chest X-ray classification uses ResNet-50 with transfer learning. The training dataset has only 5,000 labeled images. Which approach is BEST?

(A) Train ResNet-50 from scratch

(B) Freeze all layers and only train a new output layer

(C) Freeze early layers, fine-tune later layers and the output layer with a low learning rate

(D) Use a fully connected network instead of CNN

Correct Answer: (C)

Explanation: With limited data, transfer learning is essential. Freeze early layers (generic features). Fine-tune later layers (domain-specific features) with a low learning rate. Training from scratch would overfit with only 5000 images.

Question 12:

Which AWS service provides HIPAA-eligible natural language processing specifically designed for clinical text?

(A) Amazon Comprehend

(B) Amazon Comprehend Medical

(C) Amazon Textract

(D) Amazon Lex

Correct Answer: (B)

Explanation: Amazon Comprehend Medical is the HIPAA-eligible variant of Comprehend, specifically designed to extract medical entities (medications, conditions, procedures, dosages) from clinical text.

Question 13:

A SageMaker training job needs to access training data in S3 without traversing the public internet. What should be configured?

(A) S3 Transfer Acceleration

(B) S3 VPC Gateway Endpoint

(C) AWS Direct Connect

(D) CloudFront distribution

Correct Answer: (B)

Explanation: An S3 VPC Gateway Endpoint allows resources in a VPC to access S3 privately without going through the internet. This is a common HIPAA/security requirement for healthcare ML workloads.

Question 14:

A data engineer needs to build an ETL pipeline that: (1) crawls data in S3, (2) catalogs the schema, (3) transforms the data, and (4) tracks what has already been processed. Which combination of services handles ALL four requirements?

(A) AWS Glue (Crawlers + Data Catalog + ETL Jobs + Job Bookmarks)

(B) Amazon EMR + AWS Glue Data Catalog

(C) Amazon Athena + AWS Glue Crawlers

(D) AWS Lambda + Amazon S3 Events

Correct Answer: (A)

Explanation: AWS Glue provides all four: Crawlers (discover schema), Data Catalog (store metadata), ETL Jobs (transform), and Job Bookmarks (track processed data). The other options handle some but not all requirements.

Question 15:

SageMaker Automatic Model Tuning is running Bayesian optimization to find the best learning rate for an XGBoost model. After 50 trials, the best RMSE is 0.15. Should the data scientist switch to random search?

(A) Yes — random search is always better

(B) Yes — Bayesian may be stuck in a local minimum

(C) No — Bayesian optimization is more sample-efficient and will continue to improve

(D) No — but they should switch to grid search

Correct Answer: (C)

Explanation: Bayesian optimization builds a surrogate model and gets more efficient over time. After 50 trials it has a good model of the search space. Switching to random would lose this knowledge. More trials with Bayesian will yield better results.

Question 16:

A clinical trial dataset has the following class distribution: Drug Response (positive) = 8%, No Response (negative) = 92%. The current model has 93% accuracy. Which TWO approaches would MOST improve minority class detection? (Select TWO)

(A) Apply SMOTE to oversample the positive class

(B) Increase model complexity

(C) Adjust class weights to penalize false negatives more heavily

(D) Remove features to reduce overfitting

Correct Answer: (A) and (C)

Explanation: SMOTE creates synthetic minority samples to balance the dataset. Class weights increase the cost of misclassifying the minority class. Both directly address the imbalance problem. More complexity or fewer features don't address class imbalance.

Question 17:

A data scientist needs to reduce a dataset from 500 features to 50 features while preserving the maximum amount of information. Which technique should they use?

(A) t-SNE

(B) PCA

(C) K-Means

(D) Random Cut Forest

Correct Answer: (B)

Explanation: PCA finds the directions of maximum variance and projects data onto fewer dimensions. It preserves the most information (variance) possible in the reduced space. t-SNE is for visualization only (2-3D). K-Means is clustering, not dimensionality reduction.

Question 18:

Which metric is MOST appropriate to evaluate a spam filter where false positives (legitimate email marked as spam) are MORE costly than false negatives (spam reaching inbox)?

(A) Recall

(B) Precision

(C) F1 Score

(D) Accuracy

Correct Answer: (B)

Explanation: When false positives are costly, optimize for Precision = TP/(TP+FP). High precision means when the model says 'spam,' it's almost certainly spam, minimizing the chance of filtering legitimate email.

Question 19:

A model needs to be deployed to a device at a remote clinic with no reliable internet. The device has an ARM processor. Which TWO services are needed?

(A) SageMaker Neo (compile for ARM) + AWS IoT Greengrass (edge deployment)

(B) SageMaker Real-Time Endpoint + Amazon API Gateway

(C) SageMaker Batch Transform + S3

(D) SageMaker Serverless Inference + Lambda

Correct Answer: (A)

Explanation: SageMaker Neo compiles the model for ARM hardware. IoT Greengrass runs the model locally on the edge device, enabling inference without internet. All other options require cloud connectivity.

Question 20:

A SageMaker Pipeline includes: data processing → training → evaluation → conditional registration. The model should only be registered if accuracy > 0.85. Which Pipeline step type handles this?

(A) ProcessingStep

(B) TrainingStep

(C) ConditionStep

(D) TransformStep

Correct Answer: (C)

Explanation: ConditionStep evaluates conditions (like accuracy > 0.85) and branches the pipeline accordingly. It can gate model registration based on evaluation metrics.

5.3 Test-Taking Strategy (15 min)

Before Exam Day

- Review this guide's tables and cheat sheet — these cover 80% of exam content
- Take at least 2 full practice exams (65 questions each, 180 minutes)
- Focus on weak domains identified by practice exams
- Review AWS documentation for any services you're unfamiliar with
- Get hands-on with SageMaker if possible — even the free tier helps

Day of Logistics

- Exam: 65 questions, 180 minutes (2 min 46 sec per question average)

- Format: Multiple choice (1 correct) and multiple response (2+ correct)
- Passing score: 750 / 1000 (approximately 75%)
- You can flag questions and return to them
- No penalty for guessing — NEVER leave a question blank
- Scratch paper / whiteboard provided at testing center

During-Exam Question Strategy

For each question, follow this process:

1. Read the LAST sentence first (the actual question being asked)

2. Identify keywords that map to AWS services (see keyword table below)

3. Eliminate obviously wrong answers (usually 1-2 are clearly wrong)

4. Between remaining options, choose the one that is MOST operationally efficient, MOST cost-effective, or requires LEAST custom code (unless the question asks otherwise)

5. If still unsure, flag it and move on. Come back with fresh eyes.

Two-Pass Method

Pass 1: Answer all questions you're confident about. Flag uncertain ones. (Target: 45 min)

Pass 2: Return to flagged questions. With reduced pressure, you'll often see the answer clearly. (Target: 25 min)

Remaining time: Review any questions where you selected an answer that 'felt wrong.'

AWS Exam Keyword → Answer Mapping

Keyword in Question	Likely Answer
'real-time' + 'streaming'	Kinesis Data Streams
'deliver to S3' + 'near-real-time'	Kinesis Data Firehose
'serverless ETL'	AWS Glue
'SQL on S3 data'	Amazon Athena
'tabular data' + 'classification/regression'	XGBoost
'forecast' + 'multiple time series'	DeepAR
'anomaly detection' + 'streaming'	Random Cut Forest
'text classification'	BlazingText (supervised)
'discover topics'	LDA or NTM
'translate text'	Amazon Translate (or Seq2Seq if SageMaker)
'no ML expertise' + 'recommendations'	Amazon Personalize

Keyword in Question	Likely Answer
'clinical text' + 'extract entities'	Amazon Comprehend Medical
'reduce dimensions'	PCA
'compile model for edge'	SageMaker Neo

Time Management Blueprint

Phase	Questions	Time	Strategy
Easy pass	~40 questions	60 min	1.5 min each; answer and move on
Medium pass	~15 questions	45 min	3 min each; careful elimination
Hard / flagged	~10 questions	30 min	3 min each; use keyword mapping
Final review	All flagged	25 min	Re-read, check for misreads
Buffer	—	20 min	Safety margin for tough questions

Common Mistakes to Avoid

- Don't overthink: AWS exams reward knowing the 'standard' answer, not edge cases
- Don't confuse Kinesis Data Streams with Firehose — this is the #1 mix-up
- Don't assume deep learning is always the answer — XGBoost beats DL on tabular data
- Don't forget that tree-based algorithms DON'T need feature scaling
- Don't confuse precision and recall — memorize which error each minimizes
- Don't pick SageMaker for tasks where a managed AI service exists (e.g., Translate, Forecast)
- Don't ignore the 'MOST cost-effective' or 'LEAST operational overhead' qualifiers

Advice from Those Who Passed

"Focus on understanding WHEN to use each service, not HOW to configure it. The exam is about selection, not implementation." — Solutions Architect

"I made flashcards for every SageMaker algorithm with its use case and exam keywords. That alone covered 40% of the questions." — Data Engineer

"The modeling domain (36%) is where I spent most of my study time, and it paid off. Know your metrics and when to use each." — ML Engineer

"Don't underestimate the data engineering section. Kinesis questions are tricky if you haven't memorized the differences between KDS, Firehose, and Analytics." — Data Scientist

"Practice exams are essential. I took 3 full practice tests and scored 80%+ on all of them before sitting the real exam." — Cloud Architect

5.4 Closing & Next Steps (5 min)

Next Steps for Students

- Review this guide's tables and cheat sheet within 24 hours (spaced repetition)
- Take at least 2 full-length practice exams (Tutorials Dojo, Whizlabs, or AWS official)
- Focus your remaining study on your weakest domain
- Schedule the exam within 2 weeks while material is fresh
- Join the AWS Certified community on Reddit or LinkedIn for tips

Exam Day Tips

- Get a good night's sleep — cognitive performance drops significantly with poor rest
- Arrive 15 minutes early to settle in
- Use the provided scratch paper for confusion matrices and diagrams
- If stuck between two answers, go with the AWS-native / managed / serverless option
- Trust your first instinct unless you have a specific reason to change

Resources

- AWS Machine Learning Specialty Exam Guide (official PDF from AWS)
- SageMaker Developer Guide (AWS Documentation)
- AWS Well-Architected Machine Learning Lens
- Tutorials Dojo Practice Exams for MLS-C01
- Andrew Ng's Machine Learning course (foundational concepts)
- AWS re:Invent ML/AI session recordings (YouTube)

Appendix A: All Quiz Answers — Quick Reference

Domain 1: Data Engineering (8 Questions)

Q1: (B) — Firehose is purpose-built for delivery to S3/Redshift with zero custom code. KDS...

Q2: (C) — At 1 Gbps, transferring 150TB would take ~14 days over the wire. Snowball Edge (...

Q3: (C) — Job Bookmarks track which data has been processed so incremental jobs only proce...

Q4: (C) — Athena is serverless interactive SQL that queries S3 directly. Redshift requires...

Q5: (D) — RecordIO-Protobuf is optimized for SageMaker built-in algorithms and supports Pi...

Q6: (A) — IoT Core handles MQTT ingestion. KDS enables real-time processing with Lambda. F...

Q7: (C) — Kinesis Data Analytics supports SQL and Apache Flink for real-time stream proces...

Q8: (B) — The curated zone holds cleaned, validated, standardized data (typically Parquet)...

Domain 2: Exploratory Data Analysis (7 Questions)

Q1: (C) — Log transformation reduces right skew and makes the distribution more normal. Af...

Q2: (C) — 500 one-hot columns creates curse of dimensionality. Label encoding implies fals...

Q3: (B) — t-SNE is for visualization only — it doesn't preserve global structure, isn't de...

Q4: (B) — Recall = TP / (TP + FN). The model catches 0 fraud cases (TP = 0, FN = 50), so r...

Q5: (C) — MNAR: the probability of data being missing depends on the unobserved value itse...

Q6: (B) — Logistic regression is sensitive to feature scales. Income (20K-500K) would domi...

Q7: (C) — High correlation (0.97) indicates multicollinearity. Keeping both adds redundanc...

Domain 3: Modeling (10 Questions)

Q1: (B) — XGBoost is the go-to algorithm for structured/tabular data. It handles mixed fea...

Q2: (C) — DeepAR is designed for forecasting multiple related time series. It learns acros...

Q3: (B) — Pipe Mode streams data from S3 directly, reducing startup time and disk requirem...

Q4: (B) — Large gap between training (95%) and validation (60%) = overfitting. The model m...

Q5: (C) — In cancer screening, missing a true cancer case (False Negative) is far worse th...

Q6: (B) — L1 (Lasso) regularization adds |w| penalty, which has a sharp corner at zero, ca...

Q7: (C) — Softmax outputs a probability distribution across all classes (sum = 1). Sigmoid...

Q8: (C) — Bayesian optimization builds a surrogate model of the objective function, making...

Q9: (B) — Skip connections allow gradients to flow directly through shortcut paths, preven...

Q10: (B) — Model Parallelism splits the model across multiple GPUs. Data Parallelism replic...

Domain 4: ML Implementation & Operations (8 Questions)

Q1: (B) — Real-time endpoints provide millisecond latency for live applications. Batch is ...

Q2: (C) — Shadow deployment sends traffic to both models but only serves the old model's r...

Q3: (B) — Model Monitor's Data Quality monitoring compares live input data statistics agai...

Q4: (B) — Network Isolation mode prevents the training container from making ANY outbound ...

Q5: (B) — Multi-Model Endpoints host many models on a single endpoint, loading/unloading d...

Q6: (B) — Comprehend Medical is specifically designed to extract medical entities (medicat...

Q7: (C) — Different false negative rates across demographic groups is a post-training bias...

Q8: (B) — SageMaker Neo compiles models for specific hardware targets (x86, ARM, GPU, edge...

Practice Exam (20 Questions)

Q1: (A) — KDS handles the real-time stream for anomaly detection via Lambda. Firehose deli...

Q2: (C) — RecordIO-Protobuf is SageMaker's optimized format and supports Pipe Mode, which ...

Q3: (C) — Root cause: far more features (50K) than samples (200). PCA reduces dimensionali...

Q4: (B) — BlazingText supervised mode is designed for fast text classification. Seq2Seq is...

Q5: (B) — With 88% non-readmitted patients, achieving 92% accuracy is easy by mostly predi...

Q6: (B) — SageMaker Experiments is designed to organize, track, and compare training runs....

Q7: (C) — KNN uses distance calculations. Without scaling, WBC count (4000-11000) would do...

Q8: (D) — Async Inference handles large payloads and long processing times. It queues requ...

Q9: (B) — L1 (Lasso) adds |w| penalty with a sharp corner at zero, enabling automatic feat...

Q10: (C) — Data drift is the most common cause of production model degradation. The real-wo...

Q11: (C) — With limited data, transfer learning is essential. Freeze early layers (generic ...

Q12: (B) — Amazon Comprehend Medical is the HIPAA-eligible variant of Comprehend, specifica...

Q13: (B) — An S3 VPC Gateway Endpoint allows resources in a VPC to access S3 privately with...

Q14: (A) — AWS Glue provides all four: Crawlers (discover schema), Data Catalog (store meta...

Q15: (C) — Bayesian optimization builds a surrogate model and gets more efficient over time...

Q16: (A) and (C) — SMOTE creates synthetic minority samples to balance the dataset. Class weights i...

Q17: (B) — PCA finds the directions of maximum variance and projects data onto fewer dimens...

Q18: (B) — When false positives are costly, optimize for Precision = TP/(TP+FP). High preci...

Q19: (A) — SageMaker Neo compiles the model for ARM hardware. IoT Greengrass runs the model...

Q20: (C) — ConditionStep evaluates conditions (like accuracy > 0.85) and branches the pipel...

Appendix B: Life Sciences Examples Summary

Quick reference of all life sciences examples organized by domain. Use during class preparation to plan which examples to emphasize based on audience background.

Domain 1: Data Engineering

1.1 — IoT Wearable Ingestion: Pharmaceutical clinical trial with 5,000 wearable devices. IoT Core → KDS (real-time alerts) + Firehose (S3 batch). Why Firehose alone isn't enough for patient safety alerts.

1.1 — Genomics Data Migration: 2TB/run genomics lab. DataSync for ongoing transfers vs. Snowball for 200TB historical migration.

1.2 — Multi-Modal Biotech Storage: Protein structures, gene expression, microscopy images. File format decisions and data lake zone mapping.

1.2 — EHR Transformation: 50M patient records: Glue ETL for bulk transform + Lambda/Comprehend Medical for PHI de-identification.

Domain 2: Exploratory Data Analysis

2.1 — Clinical Trial EDA: Cancer drug trial: histogram (ages), box plot (tumor sizes — catch mm vs cm error), scatter plot (dosage vs response), heatmap (biomarker correlations).

2.1 — Gene Expression Distributions: Log-normal distribution of gene expression data. Log transform to achieve normality.

2.2 — Hospital Readmission Features: MNAR smoking status, high-cardinality ICD-10 codes, mixed-scale lab values. Encoding and scaling decisions.

2.2 — Drug Discovery Imbalance: 0.1% active molecules. Why 99.9% accuracy is meaningless. SMOTE, class weights, anomaly detection framing.

Domain 3: Modeling

3.1 — Algorithm Selection Walkthrough: 5 life sciences problems mapped to algorithms: chemotherapy response (XGBoost), ICU demand (DeepAR), patient clustering (K-Means), device anomalies (RCF), pathology classification (Image Classification).

3.1 — Adverse Event NLP: 500K adverse event reports: LDA/NTM for topic discovery, BlazingText for severity classification. Why Seq2Seq is wrong.

3.2 — Medical Image Training: 2M chest X-rays: GPU instances, Pipe Mode, Spot Training, Data Parallelism. Cost savings calculation.

3.3 — Cancer Screening Metrics: False Negative (missed cancer) vs False Positive (unnecessary biopsy). Optimize recall for screening, precision for spam.

3.3 — Drug Interaction Overfitting: 200 features, 500 samples: L1, PCA, dropout, transfer learning. Root cause analysis.

3.4 — Pathology Imaging Pipeline: Microscopy slides: tiling, ResNet CNN, transfer learning, activation functions, skip connections.

Domain 4: ML Implementation & Operations

4.1 — Sepsis Early Warning Deployment: Real-time endpoint, A/B testing v1/v2, edge deployment for rural clinic (Neo + Greengrass).

4.1 — Protein-Ligand Multi-Model: 200 models on one Multi-Model Endpoint. 95% cost savings.

4.2 — HIPAA Compliance Architecture: Full security stack: KMS, TLS, VPC, S3 endpoint, network isolation, IAM, Model Monitor.

4.2 — Clinical AI Bias Detection: Disparate false negative rates across racial groups. Clarify detection, remediation, ongoing monitoring.

4.2 — Comprehend Medical for Clinical NLP: Extract medical entities from physician notes. HealthLake for FHIR-based storage.